

# Multi-Area OSPF Routing Lab with 5 Routers

## Network Engineering Lab Portfolio

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## 1 Overview

This repository lab guide demonstrates the design, deployment, and verification of a 5-router Multi-Area Open Shortest Path First (OSPF) routing domain using Cisco IOS routers. The primary objective is to configure an Area 0 backbone, connect non-backbone areas through valid Area Border Routers (ABRs), verify inter-area Summary LSA propagation, and troubleshoot a misconfiguration involving an orphaned OSPF area.

## 2 Lab Prerequisites & Assumptions

- Understanding of multi-area OSPF concepts, including ABRs, Backbone Area 0, and Type 3 Summary LSAs.
- Physical interfaces and cabling are initialized and in an up/up operational state.
- Interconnecting subnets use a /30 mask (255.255.255.252).

### 3 Network Topology & Addressing Scheme

The topology consists of five Cisco routers (**R1**, **R2**, **R3**, **R4**, **R5**) interconnected linearly:

- **Area 1:** R1 and R2 (eth1/0).
- **Area 0 (Backbone):** R2 (eth1/1), R3 (eth1/0 & eth1/1), and R4 (eth1/0).
- **Area 2:** R4 (eth1/1) and R5 (eth1/0).

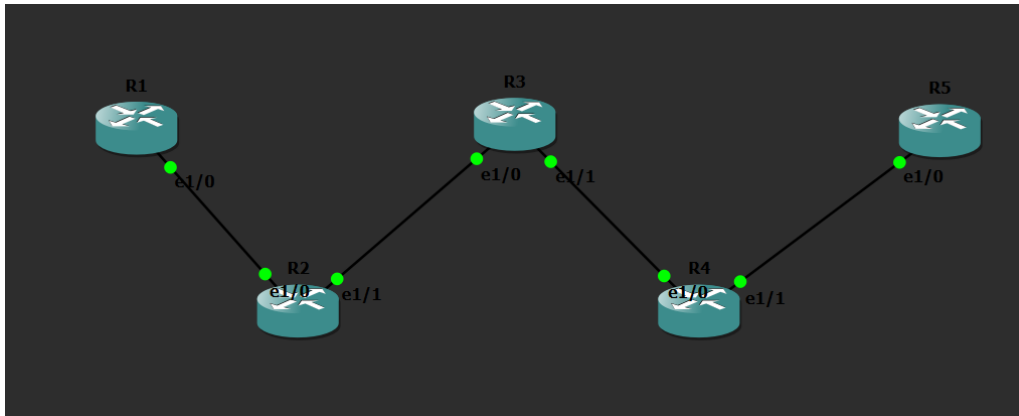


Figure 1: 5-Router Multi-Area OSPF Topology Diagram.

#### 3.1 Addressing Chart

| Router | Interface | IP Address | Subnet Mask     | OSPF Area | Role / Purpose           |
|--------|-----------|------------|-----------------|-----------|--------------------------|
| R1     | eth1/0    | 10.1.1.1   | 255.255.255.252 | Area 1    | Internal Router (Area 1) |
| R2     | eth1/0    | 10.1.1.2   | 255.255.255.252 | Area 1    | Area Border Router (ABR) |
| R2     | eth1/1    | 20.1.1.1   | 255.255.255.252 | Area 0    | Area Border Router (ABR) |
| R3     | eth1/0    | 20.1.1.2   | 255.255.255.252 | Area 0    | Backbone Transit Router  |
| R3     | eth1/1    | 30.1.1.1   | 255.255.255.252 | Area 0    | Backbone Transit Router  |
| R4     | eth1/0    | 30.1.1.2   | 255.255.255.252 | Area 0    | Area Border Router (ABR) |
| R4     | eth1/1    | 40.1.1.1   | 255.255.255.252 | Area 2    | Area Border Router (ABR) |
| R5     | eth1/0    | 40.1.1.2   | 255.255.255.252 | Area 2    | Internal Router (Area 2) |

Table 1: Lab Addressing and OSPF Area Assignment Matrix.

### 4 Step-by-Step Configuration Guide

#### 4.1 Step 1: Interface IP Address Configuration

Assign IP addresses across all interfaces and bring them operational.

**Router R1:**

```
R1(config)# interface Ethernet1/0
R1(config-if)# ip address 10.1.1.1 255.255.255.252
R1(config-if)# no shutdown
```

### **Router R2 (ABR):**

```
R2(config)# interface Ethernet1/0
R2(config-if)# ip address 10.1.1.2 255.255.255.252
R2(config-if)# no shutdown
R2(config-if)# interface Ethernet1/1
R2(config-if)# ip address 20.1.1.1 255.255.255.252
R2(config-if)# no shutdown
```

### **Router R3 (Backbone Router):**

```
R3(config)# interface Ethernet1/0
R3(config-if)# ip address 20.1.1.2 255.255.255.252
R3(config-if)# no shutdown
R3(config-if)# interface Ethernet1/1
R3(config-if)# ip address 30.1.1.1 255.255.255.252
R3(config-if)# no shutdown
```

### **Router R4 (ABR):**

```
R4(config)# interface Ethernet1/0
R4(config-if)# ip address 30.1.1.2 255.255.255.252
R4(config-if)# no shutdown
R4(config-if)# interface Ethernet1/1
R4(config-if)# ip address 40.1.1.1 255.255.255.252
R4(config-if)# no shutdown
```

### **Router R5:**

```
R5(config)# interface Ethernet1/0
R5(config-if)# ip address 40.1.1.2 255.255.255.252
R5(config-if)# no shutdown
```

## **4.2 Step 2: Multi-Area OSPF Configuration**

Enable OSPF Process 1 and assign interfaces into their respective areas using wildcard masks.

### **Router R1:**

```
R1(config)# router ospf 1
R1(config-router)# router-id 1.1.1.1
R1(config-router)# network 10.1.1.0 0.0.0.3 area 1
```

### **Router R2 (ABR):**

```
R2(config)# router ospf 1
R2(config-router)# router-id 2.2.2.2
R2(config-router)# network 10.1.1.0 0.0.0.3 area 1
R2(config-router)# network 20.1.1.0 0.0.0.3 area 0
```

### **Router R3:**

```
R3(config)# router ospf 1
R3(config-router)# router-id 3.3.3.3
R3(config-router)# network 20.1.1.0 0.0.0.3 area 0
R3(config-router)# network 30.1.1.0 0.0.0.3 area 0
```

### **Router R4 (ABR):**

```
R4(config)# router ospf 1
R4(config-router)# router-id 4.4.4.4
R4(config-router)# network 30.1.1.0 0.0.0.3 area 0
R4(config-router)# network 40.1.1.0 0.0.0.3 area 2
```

### Router R5:

```
R5(config)# router ospf 1
R5(config-router)# router-id 5.5.5.5
R5(config-router)# network 40.1.1.0 0.0.0.3 area 2
```

## 5 Troubleshooting Case Study: Isolated Area 3 Misconfiguration

### 5.1 Problem Statement & Symptoms

During the initial deployment, an area assignment mistake was made where **R4** was configured in Area 2 and **R5** was configured in Area 3 (with link 40.1.1.0/30 belonging to Area 3). Furthermore, the R3–R4 link (30.1.1.0/30) was also assigned to Area 2.

- **Symptom 1:** R5 successfully formed a FULL/DR OSPF adjacency with R4 over 40.1.1.0/30.
- **Symptom 2:** R5 was able to ping R4's local interface (40.1.1.1), but could not ping any other IP address across the topology (10.1.1.1, 20.1.1.1, or 30.1.1.1).
- **Symptom 3:** Running `show ip route ospf` on R5 returned a completely empty routing table.

### 5.2 Root Cause Analysis (RCA)

This issue occurred due to a fundamental violation of the **OSPF Hub-and-Spoke Architecture** defined in RFC 2328:

1. **Orphaned Area 3:** Area 3 was attached to Area 2 instead of connecting directly to **Area 0 (Backbone)**.
2. **Invalid ABR Status on R4:** A router is legally classified as an Area Border Router (ABR) only if it has at least one active interface inside **Area 0**. Because R4 had interfaces in Area 2 (30.1.1.2) and Area 3 (40.1.1.1), but **no link in Area 0**, R4 was not a valid ABR.
3. **Summary LSA Suppression:** Because R4 was not a valid ABR, it was prohibited from translating or forwarding Type 3 Summary LSAs (0 IA) between Area 2 and Area 3. Consequently, R5 received no inter-area routes from R1, R2, or R3.

### 5.3 Resolution Step

To resolve the issue, the topology was realigned to ensure all non-backbone areas attached directly to Area 0:

1. The transit link between R3 and R4 (30.1.1.0/30) was reassigned to **Area 0**, bringing Area 0 directly to R4's `eth1/0` interface.
2. The subnet between R4 and R5 (40.1.1.0/30) was re-captioned into **Area 2**, eliminating the isolated Area 3.

Listing 1: CLI Remediation Commands on R4 and R5

```
! On Router R4:
R4(config)# router ospf 1
R4(config-router)# no network 30.1.1.0 0.0.0.3 area 2
R4(config-router)# no network 40.1.1.0 0.0.0.3 area 3
R4(config-router)# network 30.1.1.0 0.0.0.3 area 0
R4(config-router)# network 40.1.1.0 0.0.0.3 area 2

! On Router R5:
R5(config)# router ospf 1
R5(config-router)# no network 40.1.1.0 0.0.0.3 area 3
R5(config-router)# network 40.1.1.0 0.0.0.3 area 2
```

## 6 Verification & Final Results

### 6.1 Control Plane Verification

Following the configuration correction, R4 became a legitimate ABR connected to Area 0. Inspecting the routing table on **R5** confirmed that Type 3 Inter-Area summary routes (0 IA) were successfully populated:

Listing 2: R5 Routing Table Post-Fix

```
R5#show ip route ospf
0 IA    10.1.1.0/30 [110/30] via 40.1.1.1, 00:09:12, Ethernet1/0
0 IA    20.1.1.0/30 [110/20] via 40.1.1.1, 00:09:12, Ethernet1/0
0 IA    30.1.1.0/30 [110/10] via 40.1.1.1, 00:09:12, Ethernet1/0
```

### 6.2 Data Plane Verification

End-to-end data plane connectivity across all areas was verified via ICMP pings from R5 to remote router R1:

Listing 3: R5 Ping Test Verification

```
R5#ping 10.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 24/28/32 ms
```

## 7 Conclusion

This exercise highlights a key rule in enterprise OSPF design: **every non-backbone area must have a direct connection to Area 0 via a valid ABR**. Reassigning the transit link to Area 0 established R4 as a valid ABR, enabling Type 3 Summary LSA flooding and restoring 100% network-wide reachability.